UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1 2025/2026

SECJ3553 – ARTIFICIAL INTELLIGENCE

SECTION 3

ASSIGNMENT 3: INTELLIGENT AGENT

THEME: SMART CITY

LECTURER: DR. NORSHAM BINTI IDRIS

GROUP 7

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1.0 PEAS Model

Performance	Prediction accuracy, timeliness of alert, user safety impact
Environment	Weather, river conditions, drainage status, user location, nearby facilities, historical flood records
Actuators	Notification alerts, map display, automated data visualization
Sensors	Weather APIs, river monitoring sensors, drainage sensors, GPS

2.0 PEAS Model Represented in Proof of Concept (POC)

Performance:

Our system makes use of the real-time information collected by the sensors to predict and respond to flood risks. It effectively predicts flood risk by analysing data such as rainfall intensity, humidity, river water levels, and drainage status with very high accuracy. Besides, the timeliness of flood alert is considered too. Once a high flood risk is detected, flood alerts are immediately sent to users in flood-prone areas so that users can take precautionary measures and undergo emergency rescue actions as soon as possible to reduce the impact of flood. In addition, the system enhances user safety through the Geographic Information System (GIS), Decision Support System (DSS), and the real-time communication channel. With these tools, the emergency rescue team can follow the suggested evacuation routes, keep-in-touch with the residence in affected areas to get their detailed location, and perform timely rescue operations.

Environment:

The agent operates within a smart city environment composed by physical, spatial and digital factors. In physical, the sensor will monitor the changing weather conditions, especially the rainfall intensity and humidity, alongside the river condition (the water level rise or exceed the safety limit) and the drainage status (whether the drains are blocked or flowing smoothly). Spatially, the environment involves of the user location relative to flood risk zone (red, yellow and red) and the nearby facilities needed for safe evacuation routes. Finally, the environment includes the digital archive of historical flood records, which allows the agent to compare the current environment patterns with past events to accurately predict flood risk.

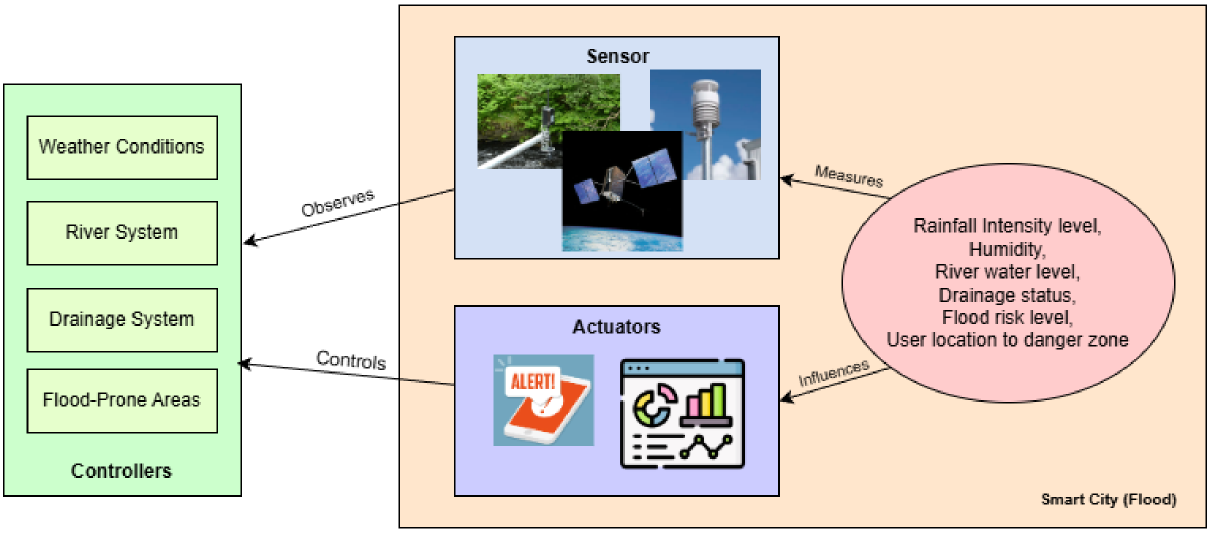
Actuators:

The actuators in this system include the flood alert system, dashboard display, and the decision support tool. Flood alerts are sent via the Short Messaging Service (SMS) and the mobile app notifications to notice users when they are in high flood risk area. This enables users to take precautionary steps as soon as possible to reduce impact of flood. Besides, the dashboard provides the visual representation of the real-time flood-risk zone map. The dashboard continuously updates the danger level of flood in each location from time to time so that users can monitor the real time flood condition in different locations. Other than that, the decision support tool suggests suitable evacuation routes, risk analysis, and response recommendations. This tool helps the emergency rescue teams to rescue flood victims in an effective and safe way.

Sensors:

To detect the environment, the smart agent utilizes a combination of physical hardware and digital interfaces. The weather application APIs act as digital sensors to continuously fetch rea-time meteorological data such as rainfall and humidity, while the physical river sensor detect the water fluctuations to ensure weather exceed the safety threshold. The drainage sensors are installed at the infrastructure device to detect blockages (is_clogged) and monitor the flow conditions. Besides, the agent integrates the GPS system as a location sensor to track the real coordinates of victims and emergency rescue teams, enabling the system to map the team to specific danger zones and coordinate accurately for rescue operations.

3.0 Relations Between Each Property



4.0 Agent in AI Behaves in App System

This project is about building an AI Flood Prediction and Early Warning System that can be considered as an intelligent utility-based agent working in a smart city environment. The ultimate purpose of the system is to offer the users an early understanding of flood situations and thus lessen the risks caused by flooding. Users by means of the mobile application, are given an opportunity to receive warnings, see flood conditions, and make good use of their time in protecting themselves when the occurrence of an emergency situation, especially in flood, prone areas in Malaysia.

The system is like a dynamic agent that keeps on communicating with the environment through the mobile application. It gets the information from various sources such as rainfall data, humidity readings, river water levels, drainage conditions, historical flood records, and user location. The data inputs enable the agent to see the changes in environmental conditions and realize the flood threats that are possible in real, time.

On launching the application by a user, the agent collects the most pertinent data automatically in accordance with the selected or current location. The system then does the processing of this data to come up with the flood risk level of the area. It is through the agent that monitoring of such signs as heavy rainfall, rising river levels, or blocked drainage systems is carried out so that the situation can be categorized into various levels of risk like low, medium, high, or severe.

In addition to keeping an eye on flood situations, the agent is offering warnings and notifications for users as well. In the case of a detection of a very dangerous flood situation, the system will send alert messages giving the details, such as location, time, and degree of severity. These notifications enable users to take preliminary steps and be ready if flooding occurs. Additionally, the live flood map in the app is changing to display the areas that are inundated in different colors so that people can easily see which are the most dangerous spots

The agent is not stopping here but is still obtaining fresh data and reevaluating its view as the change of situation. All the predictions, alerts, and map information regarding the flood are available to the public at any time and are updated regularly. Also, users are enabled to report floods through the app, thus providing assistance to the system in gaining local knowledge of areas without physical sensors.

After that, the system gradually recalibrates itself on the basis of past floods and feedback from users. It correlates historical data with present conditions to be more accurate in tracing flood patterns and forecasting risks. Therefore, the endless learning cycle is the chief factor that makes the system able to respond future floods more effectively.

In fact, the AI Flood Prediction and Early Warning System can be considered as an intelligent agent which senses changes in the environment, understands the situation, and reacts to it through alerts and graphics. The system is a great help to the community in terms of safety as it issues timely information and enables users to be up, to, date during floods.